

Unit 1, Lesson 2: Adding and Subtracting Fractions

Benchmark

MA.5.AR.1.2 Solve real-world problems involving the addition, subtraction, or multiplication of fractions, including mixed numbers and fractions greater than 1.

Learning Target

- ☐ I can solve real-world problems involving adding and subtracting fractions.

1.2.1 Warm-Up: True Statement

1. Place a digit in each blue box to make the statement true. Use only digits 1 to 9, at most one time each.

$$\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} + \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} = 1$$

1.2.2 Exploration: Framing a Piece of Art

Claudia bought a piece of art at the Winter Park Sidewalk Art Festival. She wanted to build her own frame for the piece of art. The piece of art has a height of $1\frac{1}{3}$ feet (ft) and a width of $\frac{3}{4}$ ft.

Claudia’s piece of art is shown.

- 1. Sketch a frame that Claudia could use. Label the width of the frame.



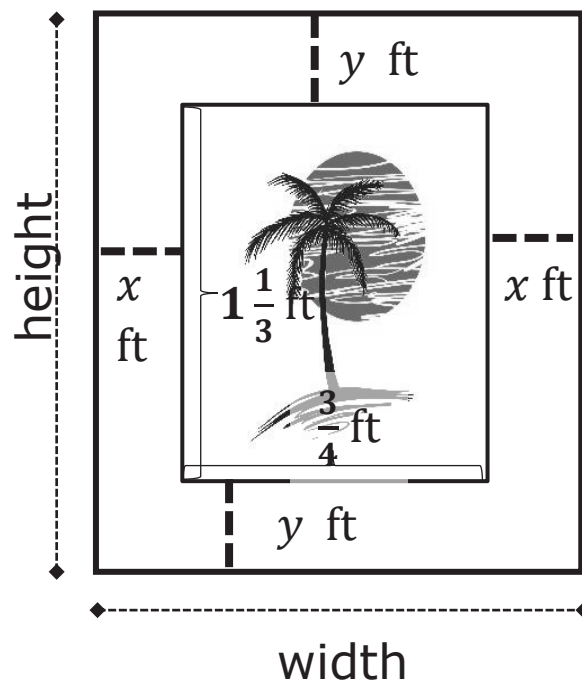
The sketches of the frames of three classmates are shown.

Eduardo	Jahmelia	Joshua

2. Which of these frames, when combined with the art piece, will produce a height that is 4 ft. taller than the original piece of art?
3. Whose frame, when combined with the art piece, will produce a width that is 1 ft. longer than the original art piece?

The frames of Eduardo, Jahmelia, and Joshua show that the width of the frame can vary. Joshua's frame shows that the frame does not have to be the same width on all four sides.

4. Claudia wants to determine the amount of wood she needs to frame the piece of art. Since she is not sure of the widths of the frame, she labels the dimensions, as shown.



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1.2.3 Guided Instruction: Adding and Subtracting Fractions with Unlike Denominators

One way to add or subtract fractions, is to first find

_____ **fractions** that have **common**, or
_____ **denominators**.

To find a common denominator for fractions, write

the

- factors
- multiples

 of the numbers in the denominators.

1. The table shows the number in each denominator of the fractions that represents the width and height of Claudia's art piece.
 - a. List at least 10 multiples of each.

3	
4	

- b. Circle the smallest multiple that both denominators have in common.
2. Working with only the fractional portion of each number, multiply both the numerator and the denominator by the factor that will result in the circled denominator.

$$\frac{1}{3} \times \frac{\quad}{\text{circled } \quad} = \frac{\quad}{\quad}$$
$$\frac{3}{4} \times \frac{\quad}{\text{circled } \quad} = \frac{\quad}{\quad}$$

Arrows point from the circled denominators in the equations above to the circled fraction in the callout box.



The circled fraction is equivalent to 1. Multiplying by 1 means that the value of the original fraction remains the same.



Multiplicative Identity Property of One

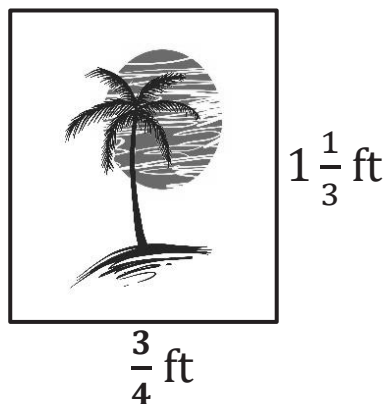
$$a \times 1 = a$$

$$1 \times a = a$$

3. Rewrite $1\frac{1}{3}$ using the equivalent fraction found for $\frac{1}{3}$ by completing the fractions.

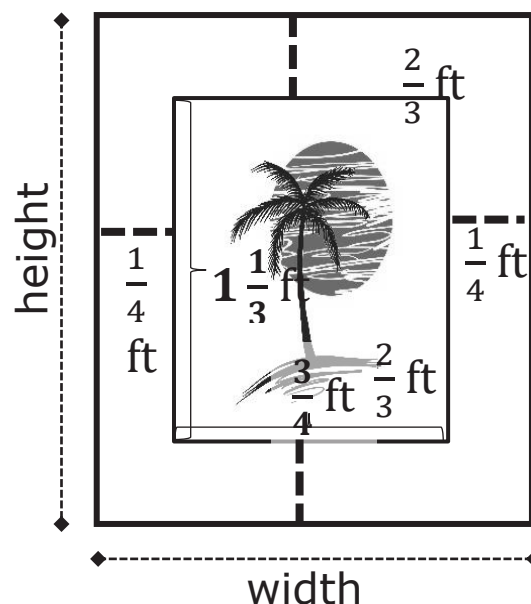
Because $1\frac{1}{3} = \frac{3}{3} + \frac{1}{3}$ then $1\frac{1}{3}$ can be rewritten using the equivalent fractions $\frac{3}{3} + \frac{1}{3} = \underline{\quad} + \underline{\quad}$.

4. Claudia's art piece is shown.



- a. What is the perimeter, in feet (ft), of Claudia's art piece?

Claudia decides to build a frame with the following dimensions.



- b. Write the numerical expression that represents the total width of the frame and art piece.

- c. Find the total width, in ft.

- d. Write the numerical expression that represents the total height of the frame and art piece.

- e. Find the total height, in ft.

1.2.4 Guided Practice: Adding and Subtracting Fractions with Unlike Denominators

Rewrite the following expressions as an expression with common denominators. Then, add or subtract the values.

1. $4\frac{1}{3} - \frac{3}{5} =$

2. $1\frac{7}{10} + 6\frac{3}{4} =$

3. $\frac{3}{5} + 1\frac{9}{10} =$

4. $2\frac{1}{6} - 1\frac{1}{4} =$

1.2.5 Wrap-Up: Ticket Out the Door

1. Use the table below to answer the questions.

Complete the table. First, find the common denominator for **Fraction A** and **Fraction B**. Then, rewrite **Fraction A** and **Fraction B** using the common denominator.

Fraction A	Fraction B	Common Denominator	Equivalent Fractions with a Common Denominator	
$\frac{7}{8}$	$\frac{5}{6}$		A:	B:
$1\frac{3}{10}$	$3\frac{2}{3}$		A:	B:
$5\frac{3}{4}$	$2\frac{1}{3}$		A:	B:

2. Find the value of each.

$\frac{7}{8} + \frac{5}{6} =$

$5\frac{3}{4} - 2\frac{1}{3} =$